**Statistics — IC0518** NAME ____________Student ID_______________

# Chapter 14.4

**Topic:** Comparing Several Multinomial Populations (Test of Homogeneity)
**Time:** ≈ 10 minutes (in pairs) **Goal:** Test, at $\alpha = 0.01$, whether the distribution of preferred portfolio risk levels differs across three income tiers.

## Scenario

An investment-advisory firm surveys **100 randomly chosen clients** from each of three income tiers and records each client's preferred portfolio risk level (Conservative, Balanced, or Aggressive). The three samples are independent. The counts are shown below.

| | Preferred Portfolio Risk Level | | | |
|---|---|---|---|---|
| **Income Tier** | **Conservative** | **Balanced** | **Aggressive** | **Total** |
| **Tier 1 (low)** | 30 | 40 | 30 | **100** |
| **Tier 2 (mid)** | 20 | 40 | 40 | **100** |
| **Tier 3 (high)** | 10 | 40 | 50 | **100** |
| **Total** | **60** | **120** | **120** | **300** |

## Tasks

1. State the null and alternative hypotheses for the test of homogeneity.
   $H_0$: ________________________________________________
   $H_a$: ________________________________________________
2. Compute the expected count $\hat{E}_{ij}$ for each of the nine cells. (Hint: $\hat{E}_{ij} = r_i \cdot c_j / n$.)
3. Fill in the work table below and compute the chi-square test statistic $X^2$.

| **Cell (i, j)** | $O_{ij}$ | $\hat{E}_{ij}$ | $O_{ij} - \hat{E}_{ij}$ | $(O_{ij} - \hat{E}_{ij})^2 / \hat{E}_{ij}$ |
|---|---|---|---|---|
| **(1, Cons)** | 30 | | | |
| **(1, Bal)** | 40 | | | |
| **(1, Agg)** | 30 | | | |
| **(2, Cons)** | 20 | | | |
| **(2, Bal)** | 40 | | | |
| **(2, Agg)** | 40 | | | |
| **(3, Cons)** | 10 | | | |
| **(3, Bal)** | 40 | | | |
| **(3, Agg)** | 50 | | | |
| | | | **$X^2$ =** | |

4. Give the degrees of freedom and the rejection region at $\alpha = 0.01$. State your conclusion.
   df = ____________  Rejection region: $X^2 >$ ____________
   Decision: □ Reject $H_0$  □ Do not reject $H_0$
   Conclusion (one sentence in the context of the scenario):
   ________________________________________________________________________

**Chi-square critical values (upper-tail).** At $\alpha = 0.01$: df = 2 → 9.210; df = 4 → 13.277; df = 6 → 16.812.

# Chapter 15.1 · The Wilcoxon Rank Sum Test (Independent Samples)

Department of Information Management & Finance, NYCU · *Time: ~10 min*

## Scenario — Comparing two robo-advisor strategies

A FinTech firm runs two robo-advisor strategies for retail clients:

**Strategy A (Growth):** a random sample of **four** monthly excess returns (%) was collected.

**Strategy B (Conservative):** an **independent** random sample of **three** monthly excess returns (%) was collected.

The analyst is unwilling to assume that monthly returns are normally distributed, so she uses the **Wilcoxon rank sum test** to ask whether the two return distributions are centered at the same level. Test at **$\alpha = 0.05$** (two-tailed).

**Data (independent samples):**

**Sample A (Growth, $n_1' = 4$):** −1, 2, 4, 5

**Sample B (Conservative, $n_2' = 3$):** 0, −3, −2

***Note:** the two samples are **independent** — the observations are not paired, and the two samples need not have the same size.*

## Tasks (work in pairs)

**Task 1 — State the hypotheses.**

Write $H_0$ and $H_a$ in words for comparing the two return distributions.

**Task 2 — Rank the combined sample.**

Pool all $\mathbf{n_1 + n_2 = 3 + 4 = 7}$ observations and rank them from smallest (rank 1) to largest (rank 7). Fill in the table below. *Tip:* the textbook convention is that *sample 1 is the smaller sample* (here B, with $n_1 = 3$); sample 2 is the larger (here A, with $n_2 = 4$).

| **Rank** | 1 | 2 | 3 | 4 | 5 | 6 | 7 |
|---|---|---|---|---|---|---|---|
| **Value** | | | | | | | |
| **Sample (A or B)** | | | | | | | |

**Task 3 — Compute the test statistic.**

(a) Let $\mathbf{T_1}$ = sum of the ranks assigned to the *smaller sample (B)*. Compute $T_1$.

(b) Compute $\mathbf{T_1^* = n_1(n_1 + n_2 + 1) - T_1}$.

(c) The test statistic is $\mathbf{T = \min(T_1, T_1^*)}$. State its value.

**Task 4 — Decision and conclusion.**

For $\mathbf{n_1 = 3, n_2 = 4}$ and a two-tailed test at $\alpha = 0.05$, Table 7 of the textbook gives the critical value $\mathbf{T = 6}$: reject $H_0$ if $\mathbf{T \le 6}$. State your decision and write a one-sentence conclusion in plain language.

## Chapter 15.2 · The Sign Test for a Paired Experiment

Department of Information Management & Finance, NYCU · *Time: ~15 min (three short linked tasks)*

### Setup — a reference distribution

Throughout this exercise, **X ~ Binomial(n = 7, p = 0.5)**. This is the null distribution used by the sign test when there are **seven** non-tied pairs. The cumulative probabilities $P(X \le k)$ are listed in the table below; you will use this table for Q1–Q3.

**Cumulative probabilities for X ~ Binomial(7, 0.5):**

| *k* | 0 | 1 | 2 | 3 | 4 | 5 | 6 | 7 |
|---|---|---|---|---|---|---|---|---|
| **P(X ≤ k)** | .008 | .062 | .227 | .500 | .773 | .938 | .992 | 1.000 |

### Q1 — From CDF to PMF.

Use the cumulative probabilities above to complete the row $P(X = k)$ below. *Hint:* $P(X = k) = P(X \le k) - P(X \le k - 1)$ for $k \ge 1$, and $P(X = 0) = P(X \le 0)$.

| *k* | 0 | 1 | 2 | 3 | 4 | 5 | 6 | 7 |
|---|---|---|---|---|---|---|---|---|
| **P(X = k)** | | | | | | | | |

### Q2 — The gourmet chef tasting (the textbook example).

Two gourmet chefs each tasted and rated eight different meals on a 1–10 scale. The ratings and the sign of *(Chef A − Chef B)* are shown below. Does it appear that one chef tends to rate higher than the other? Use **α = 0.01** (two-tailed).

| **Meal** | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 |
|---|---|---|---|---|---|---|---|---|
| **Chef A** | 6 | 4 | 1 | 8 | 2 | 4 | 9 | 7 |
| **Chef B** | 8 | 5 | 4 | 7 | 3 | 7 | 9 | 8 |
| **Sign** | − | − | − | + | − | − | 0 | − |

**Tasks:** (a) write $H_0$ and $H_a$ in terms of *p = P(Chef A > Chef B)*; (b) report the effective sample size ***n*** after dropping tied pairs and the observed value **x* = number of + signs**; (c) compute the two-tailed p-value using the table above; (d) state your decision and a one-sentence conclusion.

### Q3 — A re-tasting on different meals.

The two chefs are asked to taste and rate *eight different meals*. The new ratings and signs of *(Chef A − Chef B)* are below. Repeat the four tasks of Q2 with this new data set (use α = 0.01).

| **Meal** | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 |
|---|---|---|---|---|---|---|---|---|
| **Chef A** | 9 | 6 | 5 | 8 | 5 | 4 | 9 | 7 |
| **Chef B** | 8 | 5 | 4 | 7 | 3 | 7 | 9 | 8 |
| **Sign** | + | + | + | + | + | − | 0 | − |